When More Means Less: The Engagement Payoff of Extreme News Portrayals on Social Media

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Abstract

Biased portrayals of politicians can disproportionately attract attention on social media, potentially incentivizing news outlets to favor such content. Yet it remains unclear whether this bias–attention relationship extends beyond textual content, or whether outlets that publish biased portrayals more frequently benefit more from doing so. We analyzed the Facebook engagement of 183,516 news articles published by 48 U.S. news outlets between 2017 and 2021 and featuring eight prominent politicians. AI-assisted classifiers validated against human judgments rated headline and image portrayal from very unfavorable (1) to very favorable (5). We identify an extremity–engagement payoff: news articles that portrayed politicians with extreme stances received roughly 4 to 9 times as much engagement as those with neutral portrayals, whether favorable or unfavorable. This payoff appeared in both headlines and images but was stronger for headlines, with limited additional benefits when extreme portrayals appeared in both modalities. Although the payoff was widespread across news outlets, it tended to be smaller among outlets that published extreme headlines more frequently. These outlets also tended to have smaller, more ideologically extreme audiences and lower credibility ratings. Together, these findings reveal both the engagement incentives for extreme political news portrayals and their limits: while extremity attracts attention, outlets that rely on it more heavily do not necessarily reap greater engagement benefits.

Keywords: Media bias, visual framing, multimodal, images, social media engagement

What kinds of political news attract attention on social media, and what incentives does that attention create for news outlets? A long-standing assumption in journalism is that negative news sells. This assumption has roots in a psychological negativity bias: people attend to, process, and respond to negative information more strongly than to positive information, which shapes how they consume and react to media [1–4]. Recent work shows that these effects extend to large-scale online interaction: negative wording in news headlines increases clicks, moral-emotional language facilitates the diffusion of political messages, and posts mentioning political out-groups receive substantially more engagement than those mentioning in-groups [5–8]. Such patterns suggest that online engagement may reward news producers for publishing negative, emotionally charged, or divisive content. Media economics research suggests that outlets respond to such incentives: news content is shaped by market demand [9], including audience demand for ideologically congenial news [10]. These considerations raise a broader puzzle: if biased portrayals attract greater engagement, does publishing them more frequently necessarily yield larger engagement benefits? Understanding this relationship requires examining not only which news content attracts attention, but also how those rewards vary across different forms of presentation and news outlets.

Existing evidence offers an incomplete answer for two reasons. First, political news is not communicated through text alone. On social media, news articles commonly circulate as link previews that pair a headline with an accompanying image, and the two elements can convey different evaluations of the same politician: a headline might criticize a candidate's actions while the image presents the candidate smiling confidently. Research on these two elements, however,

has developed largely in parallel [11]. Media bias research has examined the selection, wording, and framing of political news [10, 12–16], whereas visual framing research has shown how facial expressions, body posture, camera angles, and other visual cues shape political portrayals [17–23]. Studying either modality in isolation cannot reveal how the two operate together, and there are reasons to expect them to operate differently. Headlines can be readily edited for each story, whereas images are often selected from existing photographic material [24]. Within the same news report, text tends to shape opinions, while images tend to shape behavioral intentions [25]. Because engagement is itself a behavioral response, visual portrayals may be as consequential for engagement as textual ones. Evidence on this point comes mainly from politicians' own social media posts, in which visual self-presentation, such as showing faces and displaying emotions, shapes engagement [26, 27]. News outlets, however, portray politicians differently: politicians smile more in their own posts, whereas news imagery uses more varied camera angles and close-ups [28]. Moreover, few studies have linked news images to observed audience behavior; a systematic review found that only four of 72 studies of visual or multimodal framing effects did so [29]. It therefore remains unclear whether favorable or unfavorable portrayals in headlines and images independently attract engagement, reinforce one another, or offer diminishing returns when combined.

Second, the engagement payoff of biased portrayals may depend on who publishes them and whom they portray. News outlets differ in how they portray politicians: mainstream journalism has historically emphasized professional routines and objectivity norms [30–32], whereas parts of the conservative media ecosystem developed in opposition to perceived liberal media bias [33]. These differences extend to visual portrayals; during the 2016 presidential election, for example, outlets differed in how often they depicted Hillary Clinton smiling and Donald Trump appearing angry [23]. Outlets also serve different audiences, and engagement with political news varies across partisan media environments and platforms [34–37]. Whether outlets that publish biased portrayals more frequently also gain more from them is therefore an open question, and existing theory points in two directions. Demand-side accounts of media bias hold that outlets supply the slant their audiences want [10, 38], which implies that frequent publishers of biased portrayals should benefit most. Alternatively, a strongly biased headline may stand out at an outlet that rarely publishes such content but offer little relative advantage at an outlet where such portrayals are routine. Distinguishing these possibilities bears directly on the incentive question raised above: if frequent publishers gain less from each biased portrayal, the engagement incentive to publish them may be weaker than commonly assumed. The payoff may also vary across the politicians being portrayed, since coverage of politicians differs in volume, topics, and tone [39, 40], and audience responses to coverage differ across candidates [41, 42]. How the engagement payoff of biased portrayals varies across outlets and politicians, however, has rarely been examined.

In addressing these questions, we extend research on media bias and its engagement consequences to multimodal content. We also make a methodological contribution by operationalizing media bias as stance: whether news content portrays a particular politician favorably, neutrally, or unfavorably. Large-scale studies of news engagement have relied mainly on textual cues such as negative wording, emotional intensity, and references to political in-groups or out-groups [5–7]. These measures capture the tone of content, but tone does not reveal how a particular politician is portrayed. The Fox News headline "Liberal media exploits hurricane Harvey to attack President Trump" [43], for instance, is negative in tone yet favorable toward Trump, whereas the HuffPost headline "Trump: 'This Is A Great Day' For George Floyd" [44] is positive in tone yet unfavorable toward him. Similarly, mentioning a political out-group does not necessarily mean portraying its members unfavorably. The same ambiguity applies to images, which are open to even more interpretations than text [11, 23, 29]. Although visual portrayals of politicians are often driven by facial expressions [21, 22, 45], a stern or angry face can project strength and leadership and thus be read as favorable [20, 46, 47]. Stance, by contrast, identi-

fies the target and direction of an evaluation. This is the dimension at the core of research on media bias in coverage of politicians and elections, which asks whether news favors or disfavors particular candidates [14, 23]. Measuring stance also allows us to distinguish the direction of a portrayal from its strength, or extremity (its distance from neutral), and thus to test whether engagement rewards strongly favorable as well as strongly unfavorable portrayals, beyond the established appeal of negative language.

To fill these research gaps, we examine how news headlines and images portray politicians, whether the two portrayals align, and how each relates to engagement on social media. Specifically related to potential engagement payoff and its boundary conditions, we ask two questions:

RQ1: How are the headline and image stances, separately and in combination, associated with Facebook engagement?

RQ2: How do the associations between headline and image stances and Facebook engagement differ across politicians and partisan news outlets?

Beyond understanding the incentives facing news organizations, these questions have broader implications for political communication. Social media has become a major route through which people encounter political news, often in the form of link previews rather than complete articles. According to Pew Research Center [48], 53% of U.S. adults reported getting news from social media at least sometimes, while 38% regularly obtained news from Facebook. Among regular Facebook news consumers, 85% reported receiving news from friends, family, or acquaintances. In these environments, the relative visibility of favorable and unfavorable portrayals may matter for the impressions people form of political figures [11]. This concern is particularly relevant amid growing affective polarization in the United States [49] and heightened threats and intimidation directed toward political officials [50]. Although engagement alone cannot establish changes in public attitudes or political behavior, identifying which portrayals attract disproportionate attention can help clarify one pathway through which social media news consumption may contribute to uneven and potentially biased representations of political actors.

To address the key questions, we analyzed the Facebook engagement of 183,516 news articles about eight high-profile U.S. politicians published across 48 news outlets from 2017 to 2021. For each article, we used AI-assisted classifiers validated against human judgments to rate the headline and image on a five-point scale from very unfavorable (1) to very favorable (5). Figure 1 summarizes the pipeline; full prompts, thresholds, and validation details appear in the Method section and the Supplementary Information (SI).[1]

1 Data and Method

1.1 Overview of the Pipeline

Figure 1 summarizes how we built the dataset. The pipeline has four steps. First, we selected news outlets and retrieved their articles, with headlines, image links, and Facebook engagement counts; the engagement counts provide the dependent variable. Second, we downloaded only images that contained a clear, identifiable face. Third, we identified which of these faces belonged to the target politicians. Fourth, we used a large language model to rate the stance of each headline and each image toward the pictured politician; these ratings provide the main independent variables. After merging images that showed more than one target politician and removing articles whose headline did not refer to the pictured politician or whose image the model could not match to that politician, the analytic sample comprised 183,516 articles from 48 outlets.

[1]The SI is available at `https://anonymous.4open.science/r/visual-framing-si-B4F7/SI.pdf`.

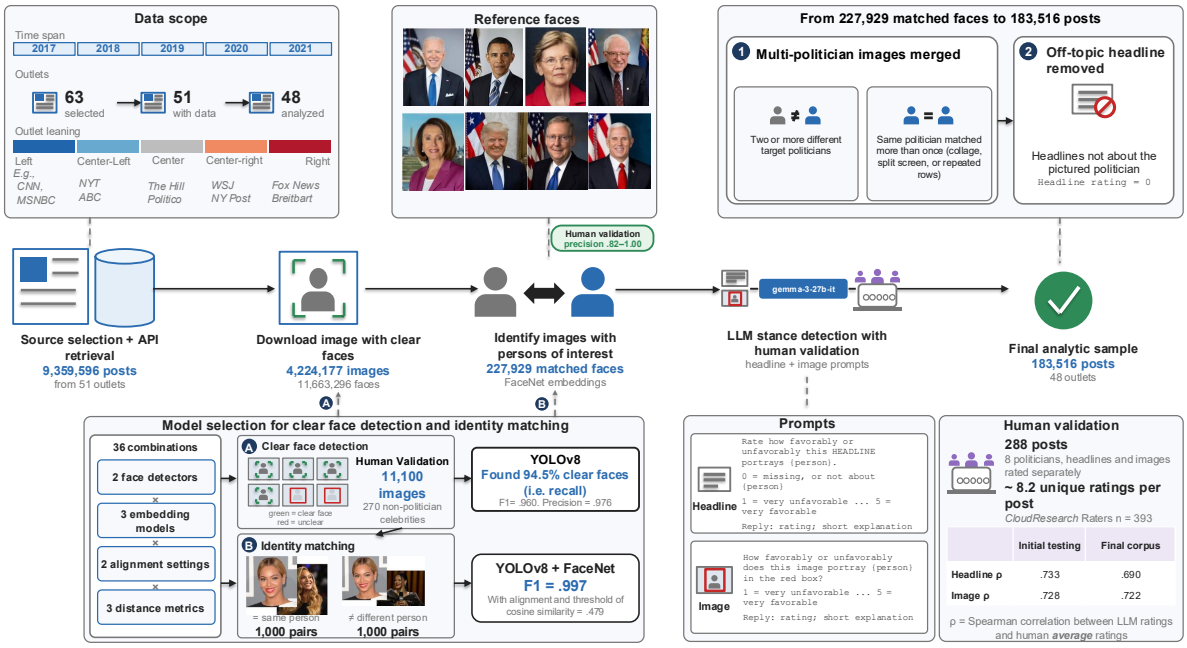


Figure 1: **Data Collection, Image Processing, and Stance Classification Pipeline.** The pipeline has four steps: (1) retrieve news articles, (2) keep images with clear faces, (3) identify images showing the target politicians, and (4) rate headline and image stance. It relies on three curated ground-truth datasets: human-coded news images of public figures other than the target politicians, same-person and different-person photograph pairs composed from those images, and a reference gallery that includes the target politicians. Human judgment checked face detection, identity matching, and stance classification; numbers give the records retained at each step. (1) NewsWhip covered 51 of the 63 outlets selected from Pew Research Center's 2014 and 2020 surveys of political news sources and from prior literature (examples shown for each leaning); the four extreme-right outlets in the analytic sample were merged with right-leaning outlets. (2) YOLOv8 found 94.5% of the faces that trained coders judged clear and identifiable (panel A). (3) Among 36 model configurations, YOLOv8 with FaceNet best separated same-person from different-person pairs ($F1 \geq .99$; panel B). Faces were then matched to the reference gallery of 1,311 public figures, and manual verification after matching gave a precision of .82 to 1.00 for the eight retained politicians. (4) gemma-4-31b-it, a free, open-weight model that larger commercial models did not significantly outperform against human judgments, rated headlines and images separately; average crowd ratings (about 8.2 raters per item) correlated with the automated ratings (Spearman $\rho = .69$–$.73$).

Because each automated step could introduce error, we checked steps 2 to 4 against human judgment, using three curated ground-truth datasets. The first is a set of 11,100 news images of 270 public figures other than the target politicians, in which trained coders judged whether each image contained a clear, identifiable face; it was used to select the face detector. The second consists of 1,000 same-person and 1,000 different-person photograph pairs composed from these images; it was used to select the detector and embedding model combination that best discriminates between individuals. The third is a hand-curated reference gallery of 11,170 photographs of 1,311 public figures, including the target politicians, against which every detected face was matched. In addition, matches to each target politician were verified manually after matching was completed, and the automated stance ratings were validated against ratings from 393 crowd participants, who rated headlines and images separately (on average 8.2 raters per headline or image).

The following sections describe each step in turn. The Measures section then defines the dependent and independent variables derived from the pipeline, along with the outlet- and article-level measures we constructed once the analytic sample was finalized; full details are provided in the SI.

1.2 Step 1: News Sources and Data Collection

We analyzed news articles published between 2017 and 2021 by major U.S. news outlets, spanning national newspapers, broadcast and cable news networks, public broadcasters, and digital-native news organizations across the political spectrum. We selected 63 outlets based on Pew Research Center's surveys of the sources Americans use for political news, namely its 2014 study of political polarization and media habits [51] and its 2020 Election News Pathways project [52], and on prior research [23, 53]. Using the NewsWhip API (accessed in summer 2024), we retrieved 9,359,596 news articles, including their headlines, image URLs, and Facebook engagement data, from the 51 outlets that NewsWhip covered. Three further outlets were excluded because their images could not be retrieved, leaving 48 outlets in the analytic sample. The outlet selection procedure and an assessment of these exclusions are described in the SI (Sections S1.2, S1.3, and S1.8). Each news article served as the unit of observation, and its Facebook engagement count provides the dependent variable (see Measures). NewsWhip records engagement for each article rather than for each Facebook post, so an article that an outlet shared more than once appears only once in the data.

1.3 Steps 2 and 3: Face Detection and Politician Identification

To identify news images featuring political figures, we developed an automated face-detection and identity-matching pipeline. Candidate models were first narrowed using published benchmarks [54]. Because these benchmarks consist mostly of frontal, well-lit photographs, whereas news images often show small, turned, or partly covered faces, we then evaluated 36 combinations of two face detectors, three embedding models, two face-alignment settings, and three distance metrics on the first two ground-truth datasets. For face detection, *YOLOv8* found 94.5% of the faces that coders judged clear and identifiable (precision = .976, $F1 = .960$). For identity matching, *YOLOv8* combined with *FaceNet* embeddings best separated same-person from different-person pairs ($F1 \geq .99$). Details of model selection and validation are provided in the SI (Section S1.4).

Using this pipeline, we downloaded 4,224,177 images containing at least one detectable face (11,663,296 faces in total) and generated a facial embedding for each face (SI, Section S1.5). We initially targeted the ten U.S. political figures most frequently mentioned in the collected headlines and matched each face against the reference gallery. Recognition precision was evaluated by manually inspecting 95–97 matched images per politician. Because precision was substantially lower for Jeff Sessions (.385) and Lindsey Graham (.271), these two individuals

were excluded. For the remaining eight politicians, precision was 1.000 for Barack Obama, .989 for Elizabeth Warren and Nancy Pelosi, .979 for Donald Trump, .958 for Mike Pence, .948 for Joe Biden, .938 for Mitch McConnell, and .823 for Bernie Sanders (SI, Section S1.6; Table S4).

Identity matching yielded 227,929 matched faces of the eight politicians. Because an image could show more than one target politician, or the same politician more than once, we merged these matches to one record per article, leaving 214,383 unique articles for stance classification (SI, Section S1.7).

1.4 Step 4: Stance Classification

Headline and image stance (see Measures) were rated with gemma-4-31b-it. For images, the model was told which politician the image was supposed to show and rated favorability toward that politician from visual cues alone; it coded 0 when it could not identify the politician in the image, which added an automated check on identity matching. For headlines, it classified stance based solely on the headline text. Headlines that did not refer to the focal politician were excluded. Of the 214,383 unique articles, 183,530 contained a headline referring to the focal politician (30,853 articles with off-topic headlines were removed), and 183,516 of these had an image in which the model identified the politician (14 images were coded 0), constituting the final analytic sample. Full model prompts and classification procedures are provided in the SI (Sections S2.2–S2.3).

1.4.1 Human Validation of Stance Measures

To assess the validity of automated stance classification, we conducted an online survey with 393 U.S. adults recruited through CloudResearch. Participants rated the favorability of a stratified sample of 288 news articles featuring the eight politicians, balanced by the politicians' party affiliations and the preliminary valence classifications of headlines and images. Headlines and images were rated separately, each shown on its own; each participant rated six headlines and six images, all from different articles. Each headline and each image received on average 8.2 independent ratings (median 8). Human favorability judgments were averaged at the stimulus level and compared with automated ratings using Spearman correlations.

During model development, automated classifications showed strong positive correlations with averaged human ratings for both headlines ($\rho = .733$, $p < .001$) and images ($\rho = .728$, $p < .001$). Agreement remained similar when the final classification procedures were reapplied to the validation stimuli after corpus-wide coding (headline: $\rho = .690$; image: $\rho = .722$; SI, Sections S2.4–S2.5).

Because favorability judgments may be shaped by partisan motivated reasoning, we tested whether raters' own partisanship colored their ratings. Democratic and Republican raters judged the same headline or image more favorably when it featured a politician of their own party (headlines: $b = 0.24$, 95% CI $[0.14, 0.35]$; images: $b = 0.25$, $[0.14, 0.36]$; both $p < .001$, on the 1–5 scale), but the effect was small ($d = 0.20$–0.23) and explained about 1% of rating variance, whereas the content being judged explained 39–45%. Democratic and Republican raters also ranked the stimuli similarly ($\rho = .60$–$.63$), and the model's ratings did not agree significantly more with either party's raters. We further benchmarked human–AI against human–human agreement. The model's ratings correlated more strongly with the consensus of the remaining raters ($\rho = .68$ for headlines, .71 for images) than a randomly drawn individual rater's ratings did ($\rho = .54$ and .48), and they reached 77% and 82% of the highest correlation attainable given the reliability of the averaged human ratings (SI, Sections S2.7–S2.8).

We also compared gemma-4-31b-it with four alternative models rated on the same validation stimuli: GPT-5.4, GPT-4.1 mini, Gemini 3 Flash, and Gemini 2.5 Flash-Lite. Using paired bootstrap tests with Holm correction, no alternative agreed with the averaged human ratings significantly better than Gemma for either headlines ($\rho = .73$–$.76$) or images ($\rho = .53$–$.76$);

Gemini 2.5 Flash-Lite agreed significantly worse on images. Given comparable validity, we chose Gemma for three reasons. First, it was available free of charge at the time of rating, whereas the commercial models are billed per token, a substantial cost for about 430,000 ratings. Second, its weights are openly released, so the ratings can be reproduced with a fixed model version after commercial versions are retired. Third, it accepts both text and images, so headline and image stance were rated by the same model with parallel prompts. Details of survey procedures, prompt development, and additional validation results are provided in the SI (Section S2).

1.5 Analytic Sample

The analytic sample comprised 183,516 articles from 48 outlets. Coverage was concentrated on the two presidents: Donald Trump appeared in 41.9% of articles and Joe Biden in 31.1%, followed by Nancy Pelosi (6.6%), Bernie Sanders (4.4%), Barack Obama (4.3%), Mitch McConnell (4.2%), Mike Pence (4.0%), and Elizabeth Warren (3.6%). Articles were split almost evenly between Democratic (49.9%) and Republican (50.1%) politicians. Of the 48 outlets, 8 (16.7%) were coded left, 14 (29.2%) center-left, 2 (4.2%) center, 4 (8.3%) center-right, and 20 (41.7%) right, including four extreme-right outlets; these groups contributed 21.4%, 30.0%, 3.8%, 9.3%, and 35.5% of articles, respectively. Most outlets were digital native (27 outlets, 56.3%), followed by newspapers (7), cable news networks (5), magazines (5), and network television (4). The number of articles per outlet was highly skewed (Mdn = 2,648, range = 29–16,292, skewness = 1.67), and the annual share of articles ranged from 13.5% in 2018 to 27.8% in 2020.

1.6 Measures

1.6.1 Facebook Engagement

The dependent variable, *Facebook engagement*, was the total number of Facebook engagements associated with each news article. NewsWhip provides an aggregate measure of Facebook engagement, combining reactions, comments, and shares without distinguishing among these interaction types. Engagement was highly right-skewed (Mdn = 679, IQR = 32–4,073; M = 8,468, SD = 48,793; skewness = 44.6), so we used log(1+engagement) in all regression analyses (M = 5.78, SD = 3.38, skewness = −0.43).

1.6.2 Headline and Image Stance

We measured stance toward political figures as the evaluative favorability expressed toward the politician depicted in an image or referenced in a headline. Both *headline stance* and *image stance* were measured on a five-point scale (1 = very unfavorable, 2 = somewhat unfavorable, 3 = neutral, 4 = somewhat favorable, 5 = very favorable).

To distinguish *stance valence* from *stance extremity*, we centered and rescaled ratings to range from −1 to 1: $H = (h - 3)/2$ for headlines and $I = (i - 3)/2$ for images, where h and i are the original five-point ratings. Positive values indicate favorable stance, negative values indicate unfavorable stance, and zero indicates neutrality. The absolute values, $|H|$ and $|I|$, measure stance extremity regardless of valence. We classified ratings of 1 or 5 as extreme. Headlines were, on average, slightly unfavorable (M = 2.68, SD = 1.02): 13.9% were rated very unfavorable, 29.6% somewhat unfavorable, 34.2% neutral, 19.9% somewhat favorable, and 2.5% very favorable. Images were closer to neutral (M = 2.86, SD = 0.99), with 1.8%, 47.7%, 17.8%, 28.4%, and 4.3% in the same five categories. Extreme ratings were more common for headlines (16.4%) than for images (6.1%).

1.6.3 Additional Outlet Characteristics

We calculated each outlet's *extreme-headline share* as the percentage of articles with headline stance rated 1 or 5 ($M = 20.2\%$, $SD = 14.3$, range = 0–59.6%; $n = 48$ outlets). Outlet *partisan leaning* was classified using Media Bias/Fact Check (MBFC) into five categories (left, center-left, center, center-right, and right, with the four extreme-right outlets merged into right), and *media type* was classified as digital native, newspaper, cable news, network television, or magazine (SI, Section S3.1).

We measured three further outlet characteristics (SI, Section S3.2; Table S6). *Audience size* was operationalized as the log-transformed average number of Facebook page followers across yearly counts from 2017 to 2021 (2018–2021 where the 2017 count was missing; $n = 48$ outlets); for Infowars, whose page was removed in August 2018, we used the 2.5 million followers reported in news coverage at the time. Follower counts were highly skewed ($Mdn = 2.43$ million, range = 0.09–31.76 million, skewness = 2.54); on the log scale they were approximately symmetric ($\log_{10}$: $M = 6.38$, $SD = 0.58$). *Credibility* was measured using NewsGuard's trust score (0–100), based on nine journalistic criteria (March 18, 2025; $M = 72.7$, $SD = 28.9$; $n = 47$ outlets, as One America News Network was not rated). *Audience ideology* was measured using the average political ideology of Facebook users sharing content from each outlet's domain, ranging from -1 (liberal) to $+1$ (conservative) [55] ($M = 0.14$, $SD = 0.67$; $n = 45$ outlets). As NewsGuard scores were collected after the study period, we treated them as approximate indicators of outlet credibility.

1.6.4 Additional Article-Level Measures

To assess whether stance–engagement associations could be explained by general emotional tone, we constructed separate measures of headline sentiment and facial expression. *Headline sentiment* was classified using a zero-shot natural language inference model (DeBERTa-v3-large-zeroshot-v2.0) on a five-point scale ranging from very negative to very positive. Sentiment scores were aggregated into a continuous measure S ranging from -1 to 1, with $|S|$ capturing sentiment magnitude (S: $M = -0.21$, $SD = 0.52$; $n = 183{,}487$ articles, as headlines from one outlet were not scored).

For facial expressions, we used DeepFace to estimate the probabilities of seven emotions (anger, disgust, fear, happiness, sadness, surprise, and neutral) for the focal politician. *Facial valence* was calculated as $F = p_{\text{happy}} - (p_{\text{angry}} + p_{\text{disgust}} + p_{\text{fear}} + p_{\text{sad}})$, with $|F|$ representing valence magnitude. Facial valence was negative on average and skewed ($M = -0.54$, $SD = 0.65$, $Mdn = -0.89$, skewness = 1.44).

Partisan congruency captures whether a portrayal matches the partisan side of the outlet that published it. It combines three pieces of information: the outlet's partisan leaning, the pictured politician's party, and whether the portrayal is favorable or unfavorable. We first coded whether the politician belongs to the outlet's own party ($G = +1$: Democratic politicians in left or center-left outlets, and Republican politicians in center-right or right outlets) or to the opposing party ($G = -1$). Interacting G with stance valence then distinguishes congruent from incongruent portrayals: $G \times H$ is positive when a headline portrays a co-partisan politician favorably or an opposing-party politician unfavorably (congruent), and negative when it portrays a co-partisan unfavorably or an opponent favorably (incongruent); $G \times I$ is defined analogously for images. Because the two center outlets have no partisan side, partisan congruency is undefined for their articles, which were excluded from models that include it (SI, Section S3.5). Among the 176,585 articles from partisan outlets, 42.3% pictured a politician of the outlet's own party; 49.8% of headline portrayals were congruent, 33.9% neutral, and 16.3% incongruent, compared with 47.3%, 17.9%, and 34.8% of image portrayals. Further measurement procedures, validation results, and alternative specifications are reported in the SI (Section S3).

1.7 Statistical Analysis

To examine how headline and image stances relate to Facebook engagement (RQ1), we estimated five ordinary least squares (OLS) models in a stepwise sequence. Models (i) and (ii) entered headline and image stance, respectively, as categorical variables, with neutral stance as the reference category, to estimate engagement at each stance level. Model (iii) interacted the five headline and five image stance categories to estimate engagement for all 25 headline–image combinations. Model (iv) replaced the categories with stance valence and extremity terms, $Y \sim (H + |H|)(I + |I|)$, separating the direction of a portrayal from its distance from neutral within and across modalities. Model (v) added partisan congruency (G, $G \times H$, $G \times I$), headline sentiment (S, $|S|$), and facial valence (F, $|F|$) to model (iv), testing whether the extremity payoff persists beyond partisan alignment and emotional tone (SI, Section S5.1). Models (i)–(iv) used all 183,516 articles from 48 outlets; model (v) used 176,556 articles from 45 outlets, because partisan congruency is undefined for the two center outlets and one outlet lacked sentiment and facial-expression scores. All models controlled for outlet partisan leaning, media type, and publication year, with standard errors clustered by outlet. The Results focus on model (v), with models (i)–(iii) describing the shape of the stance–engagement relationship (Figure 2B–D) and model (iv) serving as the stance-only comparison. Full coefficients for all five models (Tables S8–S12), additional results, and robustness checks are reported in the SI (Section S5).

To examine variation in engagement payoffs across politicians and outlets (RQ2), we estimated politician- and outlet-specific models by interacting all stance terms in model (iv) with politician or outlet indicators, respectively. Politician-specific models controlled for outlet partisan leaning, outlet type, and year, with standard errors clustered by outlet. Outlet-specific models included politician and year fixed effects, with heteroskedasticity-robust standard errors (SI, Section S6.1). The headline-extremity coefficient ($b_{|H|}$) represents the average estimated difference in log engagement between extremely favorable or unfavorable headlines and neutral headlines when image stance is neutral. The image-extremity coefficient ($b_{|I|}$) was interpreted analogously. One America News Network was excluded from outlet-specific analyses because it contributed only 29 articles, leaving 47 outlets (N = 183,487). Additional results are reported in the SI (Section S6).

We further examined whether the prevalence of extreme headlines was associated with variation in engagement payoffs across outlets. First, we regressed outlet-specific headline-extremity coefficients on each outlet's extreme-headline share (OLS with heteroskedasticity-robust HC3 standard errors). Second, we estimated article-level linear mixed-effects models with outlet random intercepts and random slopes for headline and image extremity, testing interactions between extremity and outlet-level prevalence (SI, Section S7.1). Third, to hold stable outlet characteristics fixed, we related each article's engagement to the outlet's extreme-headline share in the preceding three months, with outlet, month, and politician fixed effects and outlet- and month-specific headline-extremity slopes; this within-outlet comparison is not a causal design (SI, Section S7.8). Additional models examined moderation by outlet credibility, audience ideology, and Facebook audience size, each considered separately. Additional results and robustness checks are reported in the SI (Section S7).

2 Results

We first examined the distribution of portrayal stances in 183,516 news articles with valid headline and image ratings. Stances were rated on a five-point scale from very unfavorable (1) to very favorable (5), with neutral (3) as the midpoint. Most headlines and images conveyed neutral or moderate stances: headlines were most often neutral (34.2%), whereas images were most frequently rated somewhat unfavorable (47.7%). Overall, extreme ratings (1 or 5) were uncommon, accounting for only 16.4% of headlines and 6.1% of images. Only 1.2% of articles

contained extreme stances in both modalities. In 0.8% of articles, the two extremes shared the same valence: 0.6% portrayed politicians very unfavorably in both modalities and 0.2% very favorably in both. Nearly all of the remaining 0.4% paired a very unfavorable headline with a very favorable image. Although headline and image stance ratings were positively correlated, the association was weak (Spearman's $\rho = .122$, $p < .001$; $\chi^2(16, N = 183{,}516) = 3{,}876.52$, $p < .001$, Cramér's $V = .07$). Headlines and images thus often conveyed different evaluations of the same politician. Full distributions and cross-modal comparisons are reported in the SI (Figure S8 and Table S7).

2.1 The Extremity–Engagement Payoff Across Headlines and Images

To examine how political portrayals in headlines and images relate to Facebook engagement (RQ1), we estimated OLS regression models predicting log-transformed engagement, $\log(1 +$ Facebook engagement). We distinguished stance valence (whether a portrayal was favorable or unfavorable) from stance extremity (how far it deviated from neutral), while accounting for partisan congruency, headline sentiment, and politicians' facial expressions. The fully adjusted model, shown in Figure 2A, was specified as

$$Y = (H + |H|)(I + |I|) + G + G \times H + G \times I + S + |S| + F + |F| + \text{Controls} + \varepsilon,$$

where Y denotes log-transformed Facebook engagement; H and I denote headline and image stance valence, centered at neutral and scaled from -1 (very unfavorable) to 1 (very favorable); and $|H|$ and $|I|$ denote stance extremity, ranging from 0 (neutral) to 1 (very unfavorable or very favorable). The term $(H + |H|)(I + |I|)$ includes the main effects and all pairwise interactions between headline and image stance components. G indicates whether the politician belongs to the outlet's own party $(+1)$ or the opposing party (-1), so that $G \times H$ and $G \times I$ capture partisan-congruent portrayals. S and F denote headline sentiment and facial-expression valence, and their absolute values capture emotional intensity (SI, Sections S3.3–S3.4). All models adjusted for outlet ideology, outlet type, and year, with standard errors clustered by outlet. Because partisan congruency is undefined for center outlets, and one right-leaning outlet lacked sentiment and facial-expression scores, the fully adjusted model included 176,556 articles from 45 outlets.

The results show an extremity–engagement payoff: articles portraying politicians more favorably or more unfavorably received greater Facebook engagement, even after accounting for headline sentiment, facial expressions, and partisan congruency. Both headline extremity ($|H|$: $b = 0.80$, $SE = 0.21$, $p < .001$) and image extremity ($|I|$: $b = 0.60$, $SE = 0.12$, $p < .001$) remained positively associated with engagement. More negative headline sentiment (S: $b = -0.22$, $SE = 0.07$, $p < .001$), more intense headline sentiment ($|S|$: $b = 0.81$, $SE = 0.12$, $p < .001$), and partisan-congruent headline stance ($G \times H$: $b = 0.66$, $SE = 0.13$, $p < .001$) were also associated with greater engagement, whereas more intense facial expressions were associated with slightly lower engagement ($|F|$: $b = -0.17$, $SE = 0.06$, $p = .002$). Further analysis showed that stance and sentiment were related but distinct: headline stance and sentiment correlated moderately ($\rho = .52$, $p < .001$), but they pointed in opposite directions for 11.0% of articles. Among headlines that were negative in tone, 12.4% were favorable in stance; similarly, among headlines that were positive in tone, 20.5% were unfavorable in stance (SI, Section S3.3). Separate models estimating engagement at each level of headline or image stance (183,516 articles from 48 outlets) revealed a U-shaped pattern in both modalities. Relative to neutral headlines, very unfavorable and very favorable headlines were associated with increases of $\Delta \log = 1.46$ and 1.56, corresponding to approximately 4.3 and 4.8 times as much engagement (Figure 2B). Somewhat unfavorable and somewhat favorable headlines also received greater engagement ($\Delta \log = 0.84$ and 0.62). Images showed a similar but shallower pattern: relative to neutral images, very unfavorable and very favorable images were associated with approximately 2.1 and 1.7 times as

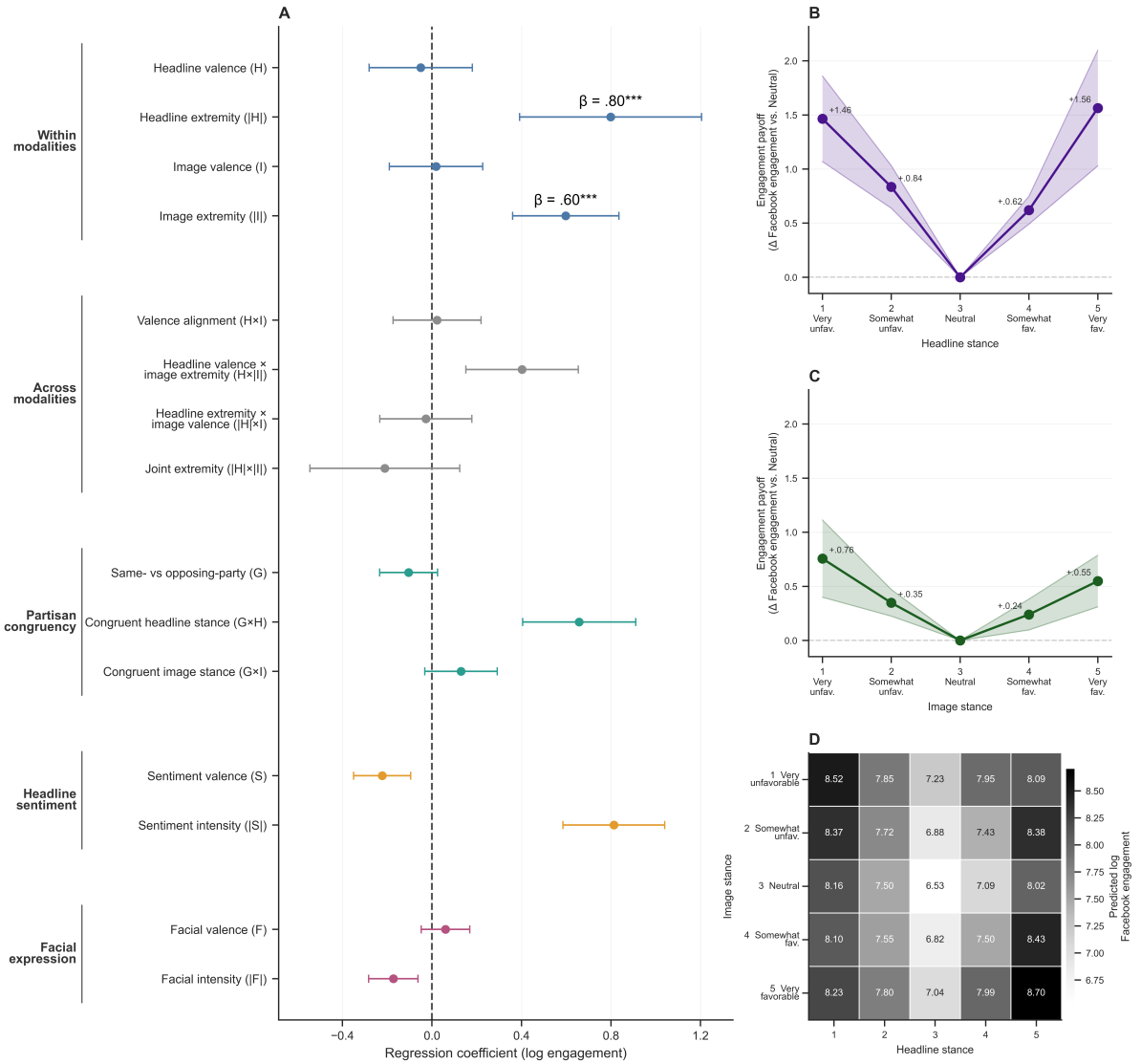


Figure 2: **The Extremity–Engagement Payoff Across Headline and Image Portrayal Stances.** Engagement is measured as log(1 + Facebook engagement). Stance ranges from very unfavorable (1) through neutral (3) to very favorable (5). Error bars and shaded bands represent 95% confidence intervals based on standard errors clustered by outlet. All models adjust for outlet ideology, outlet type, and year. **A**, Coefficients from a regression model that separates stance valence from extremity within and across modalities, and adds partisan congruency, headline sentiment, and facial expression. Within each modality, the farther the stance was from neutral, the more engagement the article received. This association was stronger for headlines than for images. Across modalities, the engagement advantage of favorable relative to unfavorable headlines increased as images became more extreme. Headlines with favorable stance toward a politician of the outlet's own party, or with unfavorable stance toward a politician of the opposing party, received more engagement. Holding stance constant, headlines with more negative sentiment and more intense sentiment received more engagement, whereas more intense facial expressions received less. The remaining terms were not significantly different from zero. **B**–**C**, Differences in log engagement relative to neutral headline stance (**B**) and neutral image stance (**C**), based on 183,516 articles from 48 outlets. Both unfavorable and favorable portrayals were associated with significantly greater engagement than neutral portrayals, with larger differences for headlines than for images. **D**, Adjusted predictions from a regression model with an interaction between categorical headline and image stance. Predicted engagement was lowest when both modalities were neutral and highest when both were very favorable. $^{***}p < .001$.

much engagement ($\Delta \log = 0.76$ and 0.55; Figure 2C). The headline payoff exceeded the image payoff in the stance-only model ($b_{|H|} - b_{|I|} = 0.82$, $SE = 0.16$, $p < .001$; SI, Section S5.5), but the difference was smaller and not statistically significant in the fully adjusted model (0.20, $SE = 0.17$, $p = .24$). The image payoff appeared to be more sensitive than the headline payoff to how the heavy-tailed distribution of engagement was modeled (e.g., Poisson pseudo-maximum likelihood; SI, Section S5.9). Both payoffs were robust to excluding any single outlet or politician, to comparing articles from the same outlet in the same week, and to comparing articles about the same politician published within a day of each other with highly similar headlines (SI, Sections S5.8–S5.11).

Extreme portrayals in both modalities were associated with the highest engagement, but evidence that headline and image valence alignment adds a payoff of its own was limited. Predicted engagement was lowest when both modalities were neutral (approximately 684 interactions, back-transformed from the log scale), compared with approximately 5,013 when both were very unfavorable and 6,002 when both were very favorable, or 7.3 and 8.8 times as much engagement (Figure 2D; joint test of headline × image interaction terms, $\chi^2(16) = 90.5$, $p < .001$). In the stance-only model, headline–image valence alignment was positively associated with engagement ($H \times I$: $b = 0.19$, $SE = 0.05$, $p < .001$), but this association was no longer significant after accounting for partisan congruency ($b = 0.02$, $SE = 0.10$, $p = .82$). One cross-modal interaction remained statistically significant: the engagement advantage of favorable over unfavorable headlines grew as images became more extreme ($H \times |I|$: $b = 0.40$, $SE = 0.13$, $p = .002$). Neither the interaction between headline extremity and image valence ($|H| \times I$) nor that between headline and image extremity ($|H| \times |I|$) was statistically significant. These results show that engagement was associated primarily with the extremity of portrayals within each modality, with limited evidence of additional payoffs from aligning headline and image valence.

2.2 More Extremity, Less Engagement Payoff

We next examined how the prevalence and engagement payoff of extreme stances varied across politicians and news outlets (RQ2). We estimated politician- and outlet-specific engagement differences between extreme and neutral stances, adjusting for the other modality and covariates. Across the eight focal politicians, extreme headlines were associated with greater engagement in every case (all $p < .001$), whereas image extremity was significantly positive for five of eight politicians. Across news outlets, the headline-extremity payoff was positive for 46 of 47 outlets (all but the *Daily Caller*, $\Delta \log = -0.84$), compared with 31 of 47 for images (SI, Section S6). Politician- and outlet-specific results for headlines and images are reported in the SI (Figures S18 and S19). The outlet-level analysis included 47 news outlets spanning five categories of political leaning. Outlet selection, political leaning classifications, and the full outlet list are provided in the Method section and the SI (Table S1).

Notably, outlets that published a greater proportion of extreme headlines showed smaller engagement payoffs (Figure 3A). Extreme-headline prevalence—the percentage of each outlet's articles with headline stance ratings of 1 or 5—ranged from approximately 2% at *PBS* and *CBS* to 59.6% at *RedState* ($Mdn = 17.5\%$). Across the 47 outlets, this share was negatively associated with the estimated engagement payoff (Spearman's $\rho = -.38$, $p = .008$). A 10-percentage-point higher share was associated with a 0.20 lower log-engagement payoff ($b = -0.20$, 95% CI $[-0.32, -0.08]$). This association held across alternative estimation approaches and was robust to excluding outlets with few articles or any single outlet (SI, Sections S7.2–S7.3). By contrast, the image-extremity payoff was unrelated to outlets' extreme-image share ($b = -0.37$ per 10 percentage points, 95% CI $[-1.07, 0.32]$, $p = .29$; SI, Section S7.2). The association also held within outlets over time: periods following a higher share of extreme headlines showed smaller headline payoffs ($b = -0.56$ per 10 percentage points, $p = .004$), although this relationship was sensitive to outlet-specific time trends and did not establish a causal effect of prevalence on

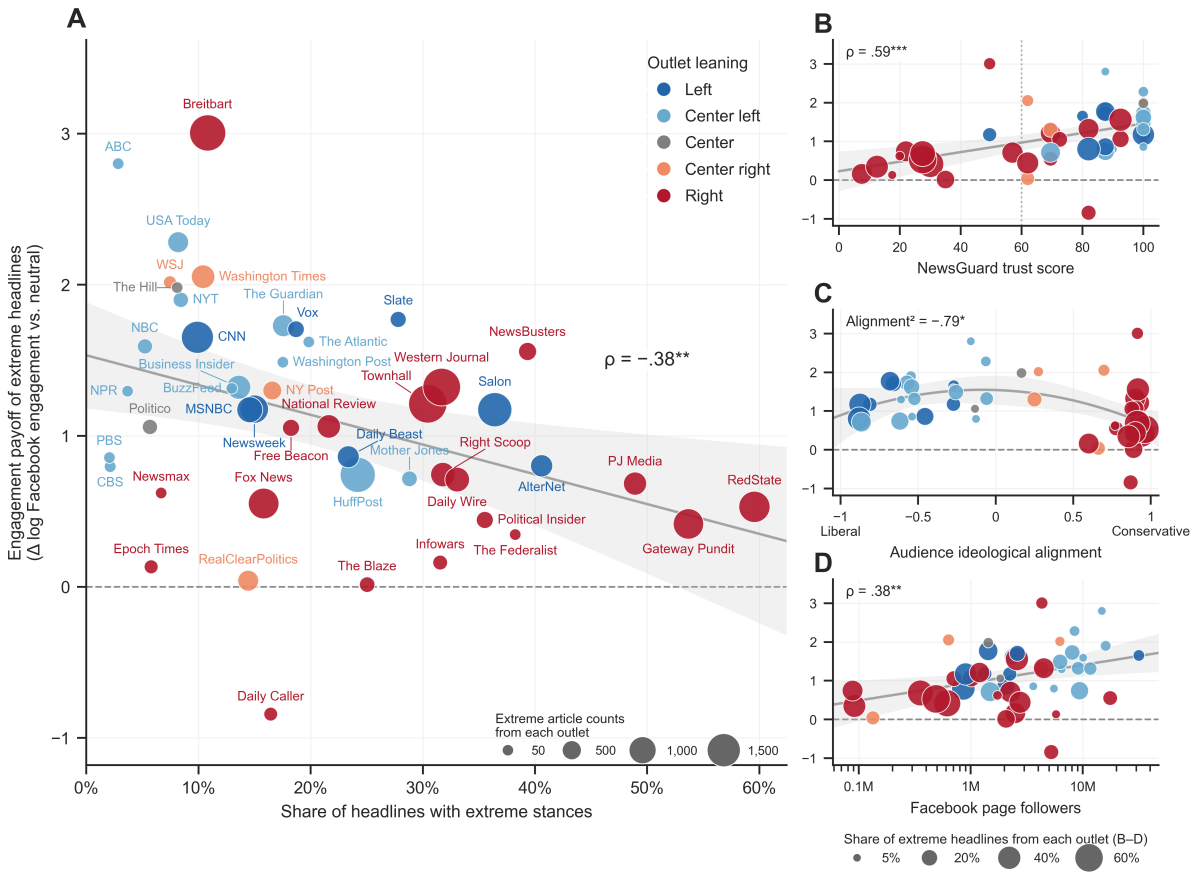


Figure 3: **Extreme-Headline Payoffs Are Larger Among Outlets That Publish Extreme Headlines Less Often, Reach More Centrist and Larger Audiences, and Are Rated More Credible.** Each point represents a news outlet, colored by Media Bias/Fact Check (MBFC) partisan leaning. The vertical axis shows each outlet's engagement payoff from extreme headlines: the difference in log Facebook engagement between extreme and neutral headline stances, adjusted for image stance and other covariates. Marker area is proportional to the number of extreme-headline articles in panel A and to the share of extreme headlines in panels B–D. **(A)** Outlets that publish extreme headlines less often receive larger payoffs from them. The horizontal axis shows the share of each outlet's articles with extreme headline stances. **(B)** Outlets rated more credible receive larger payoffs. The horizontal axis shows each outlet's NewsGuard trust score; the dotted line marks NewsGuard's credibility threshold. **(C)** Payoffs are larger for outlets whose audiences sit near the ideological center than for those at either end. The horizontal axis shows the ideological alignment of Facebook users who share each outlet's content, from liberal to conservative [55]. **(D)** Outlets with larger Facebook audiences receive larger payoffs. The horizontal axis shows each outlet's average number of Facebook page followers on a log scale. *OANN* is excluded because it had only 29 articles, none with an extreme headline, and *Daily Wire* and *Epoch Times* are further excluded from panel C for lacking audience alignment data.

engagement (SI, Section S7.8).

Outlets with higher credibility, more ideologically centrist audiences, and larger Facebook followings tended to receive larger engagement payoffs from extreme headlines. Payoffs were positively correlated with NewsGuard credibility ratings (Spearman's $\rho = .59$, $p < .001$; $b = 0.12$ per 10 points, $SE = 0.02$; $n = 47$; Figure 3B). Across audience ideology, payoffs followed an inverted-U pattern, with larger payoffs among outlets whose audiences were closer to the ideological center (quadratic term: $b = -0.79$, $SE = 0.32$, $p = .017$; $n = 45$; Figure 3C). Payoffs were also larger for outlets with more Facebook followers ($\rho = .38$, $p = .009$; $b = 0.46$ per tenfold increase, $SE = 0.15$; $n = 47$; Figure 3D). Image payoffs showed weaker associations in the same direction with credibility ($\rho = .30$, $p = .040$) and audience ideology (quadratic term: $b = -1.02$, $p = .049$), but not with audience size ($\rho = .27$, $p = .066$; SI, Sections S7.4–S7.6). Additional analyses of outlet audience size, credibility, and audience ideology, and their associations with engagement payoffs, are reported in the SI (Section S7).

These outlet characteristics were also related to how frequently outlets published extreme headlines. Outlets publishing greater shares of extreme headlines tended to have lower NewsGuard credibility scores ($\rho = -.42$, $p = .004$; $n = 47$), more ideologically extreme audiences, as measured by the absolute value of audience ideological alignment [55] ($\rho = .72$, $p < .001$; $n = 45$), and smaller Facebook followings ($\rho = -.55$, $p < .001$; $n = 47$). These patterns may reflect differences in outlets' partisan audiences, journalistic standards, and audience-building strategies.

3 Discussion

By analyzing the Facebook engagement of over 183,000 news articles over five years, we found that news taking a side was associated with greater engagement on social media. Headlines and images that portrayed a politician very favorably or very unfavorably received several times as much engagement as neutral portrayals, even after we accounted for headline sentiment, facial expression, and partisan congruency. Yet this payoff varied across outlets. Outlets that published extreme headlines most often showed the smallest payoffs, whereas the largest payoffs appeared at outlets that used extreme portrayals sparingly, were rated as more credible, reached ideologically centrist audiences, and had large followings. In short, extreme portrayals were associated with greater engagement, but this association was weakest where such portrayals were routine.

Media economics holds that outlets supply the slant their audiences demand [9, 10, 38], and engagement metrics now feed this demand back into newsrooms in near real time [56]. Our findings are consistent with an engagement premium for biased portrayals, but they also identify two boundary conditions on it. The first concerns modality: the association was stronger and more consistent for headlines than for images, and extreme portrayals in both modalities were associated with little additional engagement beyond that associated with each alone. The second concerns the outlet: the association was weaker among outlets that published extreme headlines more often. Together, these boundary conditions suggest that the engagement incentive to publish biased portrayals may be more limited than demand-side accounts alone would imply.

Why might the payoff be smaller where extreme portrayals are more common? The characteristics of the outlets with the largest payoffs point to three overlapping explanations. The most counterintuitive concerns credibility: outlets rated as more credible received larger payoffs from extreme headlines. One possible interpretation concerns what an extreme portrayal signals. An outlet with a reputation for accuracy risks that reputation each time it departs from neutrality [57], so its extreme portrayals are rarer, less expected, and arguably more informative, much as claims gain force when they come from unexpected sources [58]. At an outlet where extreme portrayals are the norm, an extreme headline is just one more. Audience composition points in the same direction. Payoffs were largest for outlets whose Facebook audiences sat near the

ideological center and smallest at both extremes. An audience near the ideological center is likely to include both readers who welcome almost any strongly favorable or unfavorable portrayal and readers who engage to object, whereas a one-sided audience that already receives a steady supply of congenial portrayals may have little further appetite for them [38, 59]. Because audience diversity is closely tied to news reliability [60], this explanation and the credibility explanation are compatible. The size of the existing audience may confer a further advantage. Outlets with more followers received larger payoffs, consistent with cumulative advantage in attention markets [61, 62]: an article that resonates with a large base of followers gathers early engagement that can trigger further sharing and wider algorithmic distribution.

We arrived at these findings by measuring bias as portrayal stance, the evaluation that news content directs at a specific politician. Research on media bias has captured slant through the selection of stories and sources, partisan phrases, tone, and the visibility of political actors [10, 12, 13, 63–65], and large-scale studies of engagement have relied on cues without a target, such as negative words, moral-emotional language, and mentions of partisan groups [5–7]. We show that stance and tone are related but not interchangeable: they correlated only moderately, pointed in opposite directions for one in nine headlines, and each predicted engagement when the other was held constant.

Consistent with prior work, we found that headlines with more negative tone received more engagement [2, 3, 5, 66]. When it came to evaluations of politicians, however, engagement was elevated in both directions: very favorable headlines received about as much engagement as very unfavorable ones, and once we separated extremity from valence, whether a portrayal was favorable or unfavorable added little. Engagement was thus associated more with how extreme a portrayal was than with whether it was negative. This pattern suggests a positivity bias in user engagement alongside the familiar negativity bias, in line with evidence that high-arousal positive content also spreads widely [67]. We also found that the headline payoff was concentrated in portrayals congruent with the outlet's side, favorable toward co-partisan politicians or unfavorable toward opponents (SI, Section S5.6). Out-group animosity is a powerful driver of engagement [7], but our results suggest that partisan audiences may also engage with unqualified praise of their own side's politicians.

Our findings also contribute to understanding how visual portrayals relate to user engagement with political news. Visual research has documented how photographs favor or disadvantage candidates [17, 23, 68–70], and studies of politicians' own social media posts have linked engagement to visual self-presentation, such as showing faces and displaying emotions [26, 27]. In news images, we found that image extremity was associated with engagement independently of the politician's facial expression, and that more intense expressions were, if anything, associated with slightly lower engagement. This suggests that image stance captures cues beyond the face, such as posture, setting, and camera angle, which measures of facial expression alone would miss.

Headline and image extremity combined largely additively, with no additional engagement associated with both being extreme, although the advantage of favorable headlines was larger when they were paired with extreme images. This pattern is consistent with experimental evidence that text and images in the same news report exert distinct rather than redundant effects [25]. The image payoff was also less consistent than the headline payoff. We see two reasons for this difference. Images are more open to interpretation than text [11, 23], so the same photograph may read as favorable to some viewers and unfavorable to others. Images are also produced differently: editors write and rewrite headlines for each story, whereas they often draw images from wire services and photo archives [24]. Because editors choose headlines story by story, headlines may respond more closely to engagement incentives, which may explain why we observed the relationship between prevalence and payoff for headlines but not for images.

Two outliers illustrate how individual politicians and outlet strategies may complicate the relationship between prevalence and payoff. Among the eight politicians, Donald Trump had the

largest headline payoff, considerably larger than Joe Biden's, even though extreme headlines were about as common for both. Trump's practice of provoking coverage across news and social media [71] may have made strongly favorable and strongly unfavorable portrayals of him especially effective at attracting attention, whereas Biden's lower-key style offered less material for such portrayals to resonate.

Among the outlets, Breitbart, the largest contributor to our sample and a central node of the right-wing media network [72], had the largest headline payoff, yet only 10.8% of its headlines were extreme, below the median outlet. Yang [24] found that Breitbart publishes at very high volume and relies heavily on content from wire services such as the Associated Press and Reuters. We speculate that this mix may help explain Breitbart's large payoff: a steady stream of conventional news keeps its extreme portrayals distinctive and lends them the appearance of mainstream reporting, in effect laundering biased content through a large volume of unbiased content. Our main results do not depend on Breitbart, but its case suggests that how outlets mix extreme and conventional content may matter for how strongly extremity is associated with engagement.

Our findings have implications beyond news engagement on social media. Extreme portrayals made up about one in six headlines, yet they received a disproportionate share of engagement, which suggests that what audiences see of politicians on social media may be more polarized than what outlets publish. Such skewed exposure may contribute to exaggerated beliefs about political opponents, which are associated with intergroup hostility [73], and to the affective polarization and normalization of hostility toward political figures that have accompanied rising threats against officials [49, 50, 74]. Because the payoff runs in both directions, uncritical celebration of allies deserves as much attention as demonization of opponents. We also found the largest payoffs at credible, mainstream outlets rather than at the partisan fringe, so the attention advantage of biased portrayals reaches mainstream news as well.

For news organizations, our findings suggest that bias sells, but within limits. Biased portrayals were associated with more engagement, but two boundary conditions caution against publishing them indiscriminately. First, the payoff was concentrated in headlines: image extremity showed smaller and less consistent payoffs, and pairing an extreme headline with an extreme image was associated with little additional engagement. Second, the payoff was smaller at outlets where extreme portrayals were routine and largest at credible outlets that used them sparingly. If credibility underlies these larger payoffs, relying on extremity more often could put that asset at risk [57], a possibility our data can raise but not test. Practitioners who track engagement metrics may therefore read them with these limits in mind: an occasional extreme portrayal was associated with a large payoff, whereas outlets that used extremity routinely showed smaller payoffs from each one.

Our study has a few limitations that directly inform future research. Our design is observational, and stance was not randomly assigned, so our data describe how payoffs differed across outlets and over a five-year period, not how an outlet's payoff would change if it published more or fewer extreme portrayals. Outlets may reserve extreme portrayals for stories that would attract attention anyway, such as scandals or crises, and although we accounted for tone, facial expression, partisan congruency, outlet, and year, unobserved features of stories may remain. The outlet characteristics associated with larger payoffs were also correlated with one another, so our data cannot fully separate the credibility, audience-composition, and audience-size explanations (SI, Section S7.7). Our engagement measure aggregates reactions, comments, and shares, so we cannot distinguish approval from outrage or identify who engaged. We also could not examine the role of the platform itself: observed engagement reflects both user choices and ranking algorithms that can amplify political content and shape exposure [75, 76]. Facebook changed how its News Feed ranked news at least twice during our study period: in 2018 it began prioritizing "meaningful social interactions" with friends and family over public content from publishers [77, 78], and after the 2020 election it temporarily adopted emergency measures that

reduced the visibility of untrustworthy news [79]. Our findings may also not generalize beyond the cases we studied. Our outlet sample overrepresents right-leaning outlets, and we examined eight high-profile politicians on one platform during a period dominated by Trump's presidency; portrayals of less prominent figures may follow different patterns. We believe stronger causal evidence will come from the field rather than the laboratory. Publishers' headline experiments offer a way to randomize stance within the same story [5, 80]. Future studies could also partner with newsrooms to test how headline and image choices relate to engagement, or use changes to platform ranking as natural experiments. Newsroom fieldwork could further show how editors weigh engagement against credibility when choosing headlines and images [32].

4 Conclusion

Social media engagement is often thought to reward negativity. By measuring bias as stance toward specific politicians in both headlines and images, we found a broader pattern: across 48 outlets and five years, strongly favorable and strongly unfavorable portrayals were both associated with several times as much engagement as neutral ones, beyond tone, facial expression, and partisan congruency. This engagement premium for biased portrayals, however, had clear boundaries. It was stronger for headlines than for images, it grew little when headline and image were both extreme, and it was smallest at the outlets that published extreme portrayals most often. For biased portrayals of politicians, more can mean less: the largest payoffs appeared where bias was the exception rather than the rule. As newsrooms increasingly monitor engagement in real time, recognizing both the appeal of biased portrayals and these boundaries matters for news organizations, for the platforms that distribute their content, and for audiences whose impressions of political leaders form in part through what attracts attention.

AI Disclosure Statement

We used Cursor, an AI-assisted code editor that gives access to large language models, in three ways. First, it helped generate and debug the Python and R code for data processing, statistical models, tables, and figures. Second, it assisted the analysis: running scripts, checking reported numbers against the analysis outputs, and implementing robustness checks that the authors specified. Third, at the final stage, it helped edit the manuscript and the supplementary information for clarity and consistency, and for agreement between the text and the analysis outputs. The authors designed the study, specified the analyses, and interpreted the results. All AI-generated code, output, and text were reviewed, verified, and revised by the authors, who take full responsibility for the content of this paper.

Separately, large language models served as research instruments, as described in the Method section and the SI. gemma-4-31b-it rated the stance of headlines and images and was validated against human ratings; GPT-4o was used to record the gender and race/ethnicity of the public figures in the face-detection benchmark; and Gemini 2.0 Flash supplied the preliminary codes used only to stratify the human-validation sample.

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